

# Boolean Support Does Not Determine Commutator Accessibility

*Direct Support, Exact Cancellation, and Low-Depth Hall Bridges*

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## Abstract

**Problem.** A generator-labelled support graph records which projected blocks  $Q_i X_g Q_j$  are nonzero, but it does not record whether two-step matrix products survive or whether their signed difference survives in a commutator. Consequently, direct support need not determine bracket accessibility or full Lie depth.

**Approach.** For a fixed orthogonal sectorization and a declared family of skew-Hermitian generators, we separate five object layers: generator support  $R_1$ , routed projected-composition support, full-word support, commutator support  $R_2$ , and cutoff-relative Lie depth  $D^{(d_{\max})}$ . We give an exact  $3 \times 3$  counterexample in which two systems have identical generator-indexed Boolean support but differ in commutator support and exact Lie depth. We then classify ordered sector pairs by the Boolean pair  $(R_1, R_2)$  and separate signed cancellation from image–kernel incidence.

**Results.** Boolean support does not determine commutator accessibility. In a centered one-dimensional model, explicit hypotheses imply a local bracket-emergent bridge, namely  $R_1(i, j) = 0$  and  $R_2(i, j) = 1$ . For matrix blocks, this conclusion requires actual product survival, rank protection, or a separate nondegeneracy certificate. A computational  $S_4$  case study exhibits ten direct channels, two bracket-emergent channels, two channels with both routed-product and full-word support but cancelled simple commutators, and seventy-six channels unreachable through the declared cutoff. Its machine-zero cancellations are computational evidence, not exact equalities.

**Boundary.** The stable contribution is a local classification of direct and bracket-emergent support together with exact cancellation and incidence mechanisms. It is not a global repair or completion theorem. A general word-level promotion theorem, full Lie-depth completion, and moving-family promotion are not established here.

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## Notation and Claim Layers

| Symbol                | Meaning                                                                  |
|-----------------------|--------------------------------------------------------------------------|
| $V$                   | finite-dimensional complex Hilbert space                                 |
| $Q_i$                 | orthogonal sector projector, $\sum_i Q_i = I$                            |
| $X$                   | declared skew-Hermitian generator                                        |
| $R_1^g(i, j)$         | direct support of generator $g$ from sector $j$ to sector $i$            |
| $R_1(i, j)$           | aggregate direct support over generators                                 |
| $C_2^{g,h,k}(i, j)$   | Boolean support of the routed projected two-step term through sector $k$ |
| $C_2^X(i, j)$         | aggregate routed-product support over $g, h, k$                          |
| $W_2^{g,h}(i, j)$     | full two-letter word support                                             |
| $W_2^X(i, j)$         | aggregate full-word support over $g, h$                                  |
| $R_2^{g,h}(i, j)$     | projected commutator support                                             |
| $R_2(i, j)$           | aggregate commutator support                                             |
| $D_{ij}^{(d_{\max})}$ | first detected Lie depth up to the declared cutoff                       |
| $D_{ij}$              | exact first Lie depth, used only when the full Lie closure is certified  |

The direction convention is fixed throughout:

$$Q_i X_g Q_j \neq 0 \iff j \longrightarrow i.$$

Within this paper, the unqualified aliases are local to the declared Lie family  $\mathcal{X} = \{X_g\}$ :

$$R_1 := R_1^{\text{Lie}}[\mathcal{X}], \quad R_2 := R_2^{\text{Lie}}[\mathcal{X}], \quad D := D_{\text{Lie}}[\mathcal{X}]$$

once the relevant Lie filtration has been declared. The associative supports  $C_2^X$  and  $W_2^X$  remain separate diagnostics.

The paper uses four claim-status levels:

1. **Theorem:** finite-dimensional algebraic statements with exact matrices;
2. **Computational Certificate:** declared  $S_4$  registrations and closure audits with reproducible thresholds and cutoffs;
3. **Computational Observation:** finite patterns without a promotion certificate, including the direct-channel product audit;
4. **Research Program:** statements explicitly identified as open and not used in the proofs.

## 1 Introduction

Once a sectorization has been fixed, several different notions of accessibility become available. They must not be collapsed into a single graph:

$$\begin{aligned} \text{direct support} &\neq \text{projected composition} \neq \text{word support}, \\ \text{word support} &\neq \text{commutator support} \neq \text{full Lie depth}. \end{aligned}$$

For represented group elements, support-graph paths can fail to survive projected matrix composition [4]. Here the operator family is different: the inputs are declared skew-Hermitian matrices  $X_g$ , and the first question is whether direct generator support determines projected commutator support.

The answer is no. This obstruction already appears in an exact three-sector counterexample. Two associative products may be individually nonzero yet equal, so their signed difference

vanishes. Changing one coefficient preserves every generator support bit while making the commutator nonzero.

The contributions are:

1. a formal separation of  $R_1$ , projected two-step terms, word support,  $R_2$ , cutoff depth, and exact Lie depth;
2. an exact same-support counterexample for commutator accessibility and depth;
3. a neutral four-class partition by  $(R_1, R_2)$ ;
4. a centered scalar proposition that produces a local bracket-emergent bridge;
5. a matrix-level incidence and rank boundary;
6. a claim-separated  $S_4$  computational case study.

No result in this paper implies that  $R_2$  restores general operator propagation or that  $(R_1, R_2)$  determines full depth; those remain separate completion questions. The low-order taxonomy can serve as input to a full Lie-depth audit, but such an audit still requires a declared filtration and closure certificate.

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## 2 Fixed Sectorized Skew-Hermitian Systems

**Definition 2.1** (Static sectorized system).

A static sectorized skew-Hermitian system is a triple

$$\mathcal{S} = (V, \{Q_i\}_{i \in I}, \{X_g\}_{g \in G}),$$

where  $V$  is finite-dimensional,

$$Q_i = Q_i^*, \quad Q_i Q_j = \delta_{ij} Q_i, \quad \sum_i Q_i = I,$$

and every  $X_g$  is skew-Hermitian:

$$X_g^* = -X_g.$$

The generators are declared input data. If they are constructed from unitary matrices  $\rho(g)$ , the logarithm branch and any numerical skew-Hermitian projection are part of the realization and must be reported. No invariant in this paper is inferred directly from  $\rho(g)$  without first fixing  $X_g$ .

### 2.1 Direct support

**Definition 2.2** (Generator-indexed and aggregate direct support).

Define

$$R_1^g(i, j) = \mathbf{1}[Q_i X_g Q_j \neq 0],$$

and

$$R_1(i, j) = \max_{g \in G} R_1^g(i, j).$$

Because the generators are skew-Hermitian, support is symmetric for each generator:

$$R_1^g(i, j) = R_1^g(j, i).$$

We nevertheless retain the ordered-pair convention because products are composed from right to left.

## 2.2 Projected products and word support

For an intermediate sector  $k$ , define the projected two-step term

$$T_{g,h,k}(i, j) = Q_i X_g Q_k X_h Q_j.$$

Its Boolean support is

$$C_2^{g,h,k}(i, j) = \mathbf{1}[T_{g,h,k}(i, j) \neq 0].$$

The aggregate routed-product support is

$$C_2^X(i, j) = \max_{g,h,k} C_2^{g,h,k}(i, j).$$

The full two-letter word block is

$$Q_i X_g X_h Q_j = \sum_k T_{g,h,k}(i, j),$$

and its support is

$$W_2^{g,h}(i, j) = \mathbf{1}[Q_i X_g X_h Q_j \neq 0].$$

Its aggregate support is

$$W_2^X(i, j) = \max_{g,h} W_2^{g,h}(i, j).$$

Thus a Boolean path in  $R_1$  is only a candidate for  $C_2$ ; a nonzero  $C_2$  term does not by itself determine  $W_2$ , because different intermediate terms may cancel.

## 2.3 Commutator support

**Definition 2.3** (Generator-pair and aggregate commutator support).

Fix an ordering of the finite generator labels. For  $g < h$ , define

$$R_2^{g,h}(i, j) = \mathbf{1}[Q_i [X_g, X_h] Q_j \neq 0],$$

and

$$R_2(i, j) = \max_{g < h} R_2^{g,h}(i, j).$$

The exact expansion is

$$Q_i [X_g, X_h] Q_j = \sum_k (T_{g,h,k}(i, j) - T_{h,g,k}(i, j)).$$

Consequently,  $R_2$  is not a Boolean function of  $R_1$ . It is also not the same object as word support: it records the signed difference of two word orders.

## 2.4 Cutoff depth and exact depth

Let

$$\mathcal{L}_{\leq 0} = \text{span}\{X_g : g \in G\},$$

and recursively

$$\mathcal{L}_{\leq d} = \mathcal{L}_{\leq d-1} + \text{span}\{[X_g, Y] : g \in G, Y \in \mathcal{L}_{\leq d-1}\}.$$

**Definition 2.4** (Cutoff-relative Lie depth).

For a declared cutoff  $d_{\max}$ , define

$$D_{ij}^{(d_{\max})} = \min\{d \leq d_{\max} : Q_i Y Q_j \neq 0 \text{ for some } Y \in \mathcal{L}_{\leq d}\}.$$

If no such  $d$  is found, write

$$D_{ij}^{(d_{\max})} = \text{unreached}.$$

The implementation may serialize this state as **999**; that integer is not mathematical infinity.

If an exact or separately certified closure  $\mathcal{L} = \text{Lie}\langle X_g \rangle$  is available, define the exact depth  $D_{ij}$  by the same minimum over the full filtration and set  $D_{ij} = \infty$  only when

$$Q_i \mathcal{L} Q_j = \{0\}.$$

## 3 Low-Order Support Partition

**Theorem 3.1** (Four low-order support classes).

*Every ordered off-diagonal sector pair belongs to exactly one of the following classes:*

| Class                        | $R_1(i, j)$ | $R_2(i, j)$ | Meaning                                                |
|------------------------------|-------------|-------------|--------------------------------------------------------|
| direct-and-bracket supported | 1           | 1           | visible at both declared low-order layers              |
| direct-only at bracket layer | 1           | 0           | directly visible; no projected commutator survives     |
| bracket-emergent             | 0           | 1           | absent from direct support and visible in a commutator |
| unresolved at layers 1–2     | 0           | 0           | absent from both low-order support layers              |

**Proof.** The ordered pair  $(R_1(i, j), R_2(i, j))$  belongs to  $\{0, 1\}^2$ , whose four elements are disjoint and exhaustive.

This theorem is a classification of low-order support status, not a depth or completion theorem. In particular,

$$R_1(i, j) = R_2(i, j) = 0$$

does not imply that  $(i, j)$  is frozen, inaccessible, or invisible to higher Hall layers.

### 3.1 Mechanisms inside the partition

The Boolean pair does not identify why a commutator vanishes. Two mechanisms must be separated.

**Cancellation mechanism.** At least one projected product term is nonzero, but every relevant signed commutator sum vanishes.

**Incidence mechanism.** A support path has nonzero block factors

$$A = Q_i X_g Q_k \neq 0, \quad B = Q_k X_h Q_j \neq 0,$$

while the product is zero:

$$AB = 0.$$

Equivalently,

$$\text{im}(B) \subseteq \ker(A).$$

A third possibility is absence of any two-step support candidate. These are mechanism labels, not additional values of  $(R_1, R_2)$  and not moving-wall categories.

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## 4 Exact Same-Support Counterexample

Let  $V = \mathbb{C}^3$  with coordinate projectors  $Q_1, Q_2, Q_3$ . Define the real skew-symmetric, hence skew-Hermitian, matrices

$$X = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \quad Y_0 = 2X,$$

and

$$Y_1 = \begin{pmatrix} 0 & 2 & 0 \\ -2 & 0 & 3 \\ 0 & -3 & 0 \end{pmatrix}.$$

**Theorem 4.1** (Boolean support does not determine commutator accessibility).

*The systems*

$$\mathcal{S}_0 = (\mathbb{C}^3, \{Q_i\}, \{X, Y_0\}), \quad \mathcal{S}_1 = (\mathbb{C}^3, \{Q_i\}, \{X, Y_1\})$$

*have identical generator-indexed direct support, but their commutator support differs. More precisely,*

$$Q_1[X, Y_0]Q_3 = 0, \quad Q_1[X, Y_1]Q_3 \neq 0.$$

*The exact Lie depth of the channel  $3 \rightarrow 1$  is infinite in  $\mathcal{S}_0$  and 1 in  $\mathcal{S}_1$ .*

**Proof.** The matrices  $Y_0$  and  $Y_1$  have the same zero pattern as  $X$ : both connect sectors 1 and 2, and sectors 2 and 3, with no direct 1–3 block. Hence the labelled  $R_1$  tensors agree in the two systems.

Since  $Y_0 = 2X$ ,

$$[X, Y_0] = 0.$$

Moreover, the Lie algebra generated by  $X$  and  $Y_0$  is the one-dimensional space  $\text{span}\{X\}$ , whose  $Q_1$ - $Q_3$  block is zero. Thus the exact channel  $3 \rightarrow 1$  is absent at every Lie depth.

Direct multiplication gives

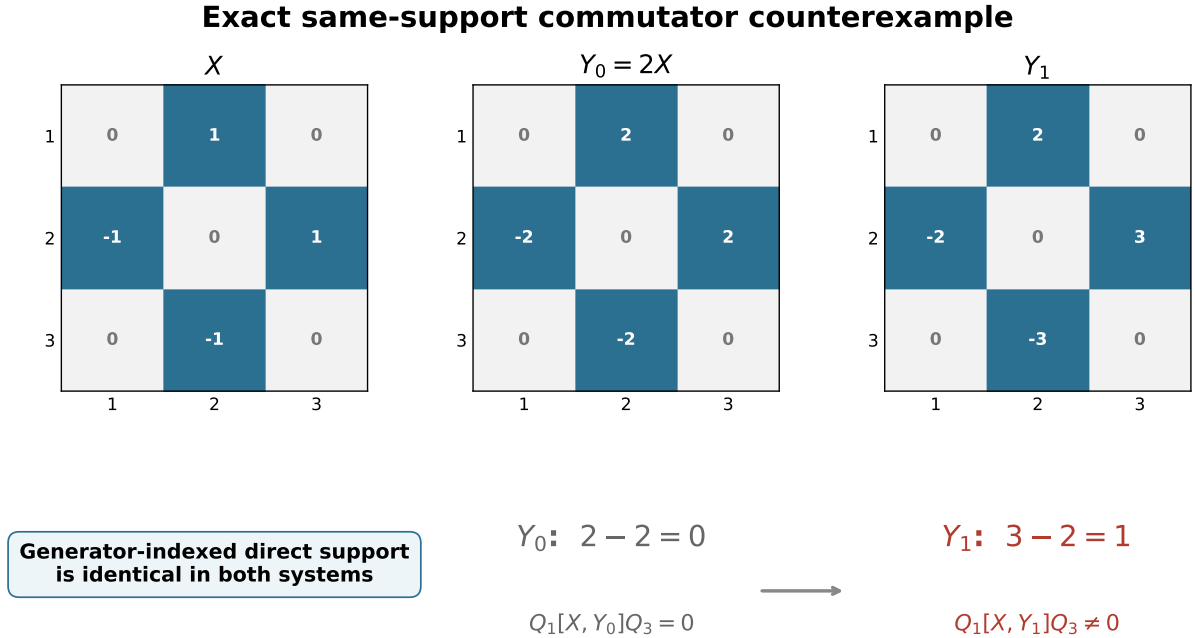
$$[X, Y_1] = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

Therefore  $Q_1[X, Y_1]Q_3 \neq 0$ , while the direct  $Q_1XQ_3$  and  $Q_1Y_1Q_3$  blocks remain zero. The channel is bracket-emergent at depth 1.

At the intermediate sector 2, the cancellation in  $\mathcal{S}_0$  is visible termwise:

$$Q_1XQ_2Y_0Q_3 = Q_1Y_0Q_2XQ_3 \neq 0.$$

This exact counterexample establishes the general negative result without relying on floating-point reconstruction or a specific group realization.



Boolean direct support does not determine signed commutator survival or exact channel depth.

**Figure 1.** The exact same-support counterexample. The two systems have identical generator-indexed direct support, but the two projected product orders cancel for  $Y_0$  and leave a nonzero commutator block for  $Y_1$ . The figure summarizes the displayed integer matrices and does not introduce a numerical claim.

## 5 Centered Scalar Bracket Emergence

The following statement isolates a sufficient condition for a local bracket-emergent bridge.

**Proposition 5.1** (Centered scalar bracket-emergence).

Assume all sectors are one-dimensional. Fix distinct generators  $g, h$ , a source sector  $j$ , a target sector  $i$ , and an intermediate sector  $k$ . Suppose:

1.  $R_1(i, j) = 0$ ;
2.  $Q_i X_g Q_k \neq 0$  and  $Q_k X_h Q_j \neq 0$ ;
3.  $Q_i X_h Q_k = 0$ ;
4. for every  $\ell \neq k$ , both  $Q_i X_g Q_\ell X_h Q_j$  and  $Q_i X_h Q_\ell X_g Q_j$  vanish;

Then

$$Q_i [X_g, X_h] Q_j \neq 0,$$

so  $(i, j)$  is bracket-emergent:

$$R_1(i, j) = 0, \quad R_2(i, j) = 1.$$

**Proof.** The sector expansion and hypotheses give

$$Q_i [X_g, X_h] Q_j = Q_i X_g Q_k X_h Q_j.$$

In one dimension the two displayed factors are nonzero scalars, so their product is nonzero.

This proposition proves emergence at the commutator layer only. It does not assert global propagation, operator-word completion, or the recovery of full Lie depth.

### 5.1 Support-set mechanism

One way to produce the hypotheses is to fix a distinguished center sector 0 and define

$$A_g = \{u : Q_u X_g Q_0 \neq 0\}, \quad B_g = \{u : Q_1 X_g Q_u \neq 0\}.$$

The conditions

$$A_g \cap B_g = \emptyset, \quad A_g \cap B_h \neq \emptyset \quad (g \neq h)$$

imply  $A_g \not\subseteq A_h$  for every ordered pair  $g \neq h$ : otherwise a point of  $A_g \cap B_h$  would lie in  $A_h \cap B_h$ . Under the additional centered-support and direct-absence hypotheses of Proposition 5.1, these set relations identify candidate bracket-emergent pairs. They do not by themselves prove matrix-product survival.

## 6 Matrix Products and Incidence

For higher-dimensional sectors, individually nonzero factors do not imply a nonzero product:

$$A \neq 0, \quad B \neq 0 \not\Rightarrow AB \neq 0.$$

**Lemma 6.1** (Image–kernel criterion).

For composable finite-dimensional linear maps  $A$  and  $B$ ,

$$AB = 0 \iff \text{im}(B) \subseteq \ker(A).$$

**Proof.** The equality  $AB = 0$  holds exactly when  $A$  annihilates every vector in the image of  $B$ .

**Lemma 6.2** (Rank protection).

If  $A$  has full column rank and  $B \neq 0$ , then  $AB \neq 0$ . Dually, if  $B$  has full row rank and  $A \neq 0$ , then  $AB \neq 0$ .

**Proof.** Full column rank gives  $\ker A = \{0\}$ , so Lemma 6.1 would force  $B = 0$  if  $AB = 0$ . Full row rank gives  $\operatorname{im} B$  equal to the entire intermediate space, so  $AB = 0$  would force  $A = 0$ .

## 6.1 Free block model versus skew-Hermitian locus

Let  $\mathcal{M}_{\text{free}}(G, \mathbf{d})$  be the affine space in which allowed complex blocks vary independently and forbidden blocks are zero. Its open support stratum requires every allowed block to be nonzero.

**Proposition 6.3** (Conditional exceptional sets in the free block model).

Fix a projected commutator channel in  $\mathcal{M}_{\text{free}}(G, \mathbf{d})$ . If its matrix-valued polynomial is not identically zero, its vanishing set is a proper algebraic subset. The region where at least one projected product survives but the commutator cancels is a constructible subset of that zero set. Similarly, the region where specified nonzero factors satisfy  $AB = 0$  is an incidence subset defined by polynomial product equations alongside open nonzero conditions.

**Proof.** Matrix products and commutators have polynomial entries in the free block coordinates. A nonzero polynomial cannot vanish on the entire affine space. Adding nonvanishing conditions removes algebraic subvarieties and hence produces constructible subsets.

This proposition applies only to the free complex-block model. Define the skew-Hermitian realization locus by

$$\mathcal{M}_{\text{skew}} \subset \mathcal{M}_{\text{free}}.$$

This locus has adjoint constraints coupling opposite blocks and is naturally a real algebraic locus. Genericity or codimension after intersection with  $\mathcal{M}_{\text{skew}}$  requires a separate argument. Standard matrix and incidence background is given in [8, 7, 5].

## 7 Computational S4 Case Study

The  $S_4$  realization supplies evidence for the low-order mechanisms and is not used in the exact proofs. It is constructed from the regular representation using three declared permutations. The numerical generators are

$$X_g = \frac{\log m(\rho(g)) - \log m(\rho(g))^*}{2},$$

with SciPy’s numerical matrix logarithm. This branch-dependent construction is part of the declared computational realization.

For this section only, the short symbols mean

$$R_1 := R_1^{\text{Lie}}[X], \quad R_2 := R_2^{\text{Lie}}[X], \quad C_2 := C_2^X, \quad W_2 := W_2^X, \quad D := D_{\text{Lie}}[X].$$

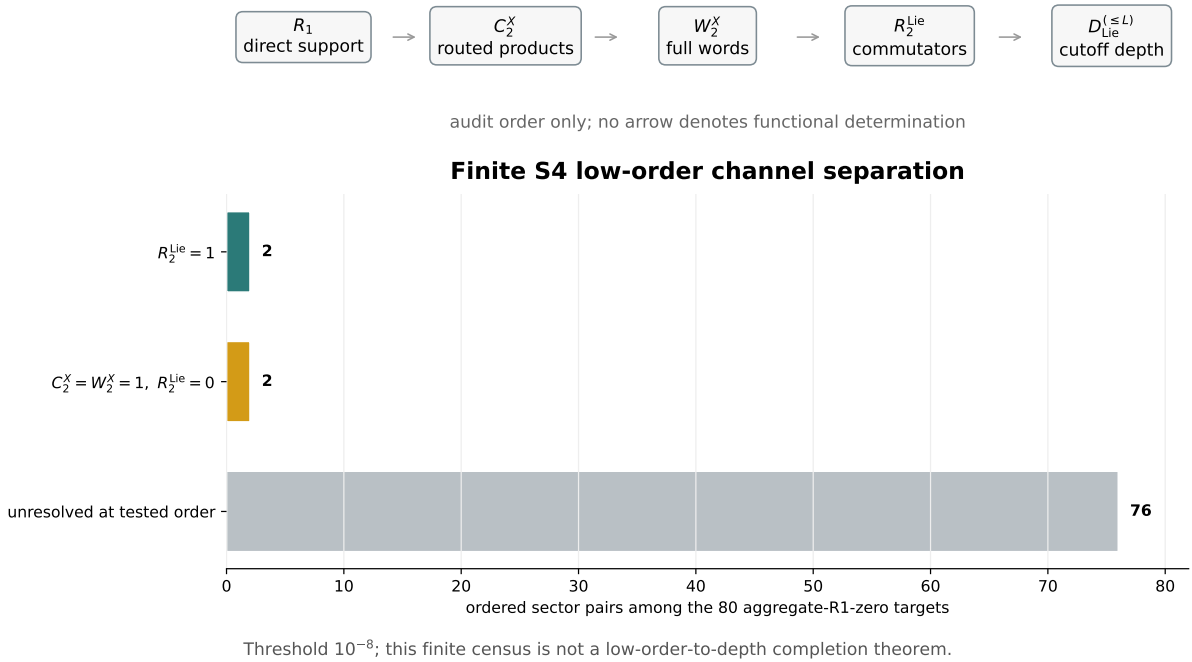
The ten sectors are produced numerically in the 24-dimensional regular representation by an ordered compression procedure. Central class sums are applied first, followed by three generator-derived Hermitian operators in the declared order. The latter do not commute: the maximum pairwise commutator norm in the seven-operator registration family is approximately 26.319. Thus this procedure defines an order-dependent orthogonal carrier decomposition, not a joint spectral resolution. The compression clustering tolerance is  $10^{-8}$ , and the resulting projectors are fixed throughout the case study. Their registration audit is:

| Registration quantity                   | Result                      |
|-----------------------------------------|-----------------------------|
| numerical dtype                         | complex128                  |
| projector ranks                         | (1, 1, 3, 3, 3, 3, 3, 2, 2) |
| maximum idempotence residual            | $3.127 \times 10^{-15}$     |
| maximum Hermiticity residual            | $1.335 \times 10^{-16}$     |
| maximum pairwise-orthogonality residual | $2.105 \times 10^{-15}$     |
| completeness residual                   | $8.697 \times 10^{-15}$     |

All residuals use the Frobenius norm. This is a numerical orthogonal-complete sector registration, not an exact representation-theoretic decomposition.

The support threshold is  $10^{-8}$ . With ten numerical sectors, the ordered off-diagonal census is:

| Low-order status                                                          | Count |
|---------------------------------------------------------------------------|-------|
| direct, $R_1^{\text{Lie}} = 1$                                            | 10    |
| bracket-emergent, $R_1^{\text{Lie}} = 0, R_2^{\text{Lie}} = 1$            | 2     |
| $C_2^X = W_2^X = 1$ but $R_1^{\text{Lie}} = R_2^{\text{Lie}} = 0$         | 2     |
| $C_2^X = W_2^X = R_2^{\text{Lie}} = 0$ among $R_1^{\text{Lie}} = 0$ pairs | 76    |



**Figure 2.** Low-order object separation in the declared  $S_4$  realization. The arrows indicate audit order only: direct support, routed products, full words, commutator support, and cutoff depth are distinct typed objects. The bar census partitions the 80 aggregate- $R_1$ -zero ordered pairs at threshold  $10^{-8}$  and does not assert low-order-to-depth completion.

The two routed- and full-word-supported pairs are  $S_4 \rightarrow S_3$  and  $S_3 \rightarrow S_4$ . For each one, the audit finds both  $C_2^X = 1$  and  $W_2^X = 1$ , while  $R_2^{\text{Lie}} = 0$ . Their two word-order norms are approximately 2.01462, while the projected commutator residuals are approximately  $6.1 \times 10^{-15}$ . These are machine-zero observations under the declared tolerance, not exact identities.

For cutoff  $d_{\text{max}} = 3$ , the numerical depth census is

$$(A_0, A_1, A_2, A_{\text{unreached}}) = (10, 2, 2, 76).$$

Here `unreached` means no block was detected through depth 3. The filtration uses generators at Lie depth 0, simple commutators at depth 1, and nested commutators at depth 2 and above. The two entries counted by  $A_2$  are exactly the two cancellation channels  $S_4 \leftrightarrow S_3$ ; they first appear through nested commutators, not through simple-commutator support.

The numerical filtration and saturation audit gives:

| Lie round | new dimension | cumulative dimension |
|-----------|---------------|----------------------|
| 0         | 3             | 3                    |
| 1         | 3             | 6                    |
| 2         | 7             | 13                   |
| 3         | 8             | 21                   |
| 4         | 0             | 21                   |
| 5         | 0             | 21                   |

At absolute rank threshold  $10^{-8}$ , the matrix augmented by the 21 retained basis vectors and all  $[X_g, L_a]$  columns has numerical rank 21. Its minimum retained singular value is 1.000, its maximum discarded singular value is  $1.400 \times 10^{-13}$ , the orthonormal-basis Gram residual is  $2.133 \times 10^{-14}$ , and

$$\max_{g,a} \text{dist}([X_g, L_a], \text{span } \mathcal{L}) = 1.344 \times 10^{-13}.$$

This supports numerical Lie-span saturation for the declared realization. It remains a computational closure certificate rather than an exact symbolic proof.

## 7.1 Direct-channel product audit

The 48-product matrix audit covers only target pairs with  $R_1(i, j) = 1$ . In those already-direct channels:

| Quantity                                        | Result |
|-------------------------------------------------|--------|
| tested single-term two-step products            | 48     |
| targets with direct $R_1$ support               | 48/48  |
| targets with $R_1 = 0$                          | 0/48   |
| zero products at tolerance $10^{-8}$            | 0/48   |
| rank-protected products                         | 48/48  |
| left-protected, $\text{rank}(A) = d_k$          | 48/48  |
| right-protected, $\text{rank}(B) = d_k$         | 48/48  |
| protected on both sides                         | 48/48  |
| sampled perturbation trials with a zero product | 0/100  |

Here  $A : V_k \rightarrow V_i$  and  $B : V_j \rightarrow V_k$ . The implemented criterion is

$$\text{rank}(A) = d_k \quad \text{or} \quad \text{rank}(B) = d_k,$$

which is exactly the full-column/full-row condition of Lemma 6.2. It is not the insufficient condition that only the smaller matrix dimension be attained.

This is a **Direct-Channel Product Nondegeneracy Audit**. It does not certify matrix-level bracket emergence, absence of incidence on  $R_1 = 0$  pairs, or  $(R_1, R_2) \rightarrow D$  completion.

## 8 Claim Status and Boundary

The object distinctions in Section 2 are definitions, not an evidence level. The table uses the four claim-status levels; proposition and lemma are result types inside the Theorem level.

| Claim                                                                                                                                      | Status                       |
|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Four classes partition ordered pairs by $(R_1, R_2)$                                                                                       | Theorem                      |
| Boolean generator support does not determine commutator support                                                                            | Theorem                      |
| Boolean generator support does not determine exact Lie depth                                                                               | Theorem                      |
| Centered one-dimensional hypotheses imply a bracket-emergent bridge                                                                        | Theorem                      |
| $AB = 0$ iff $\text{im } B \subseteq \ker A$                                                                                               | Theorem                      |
| Specified cancellation/incidence conditions define constructible free-model loci;<br>cancellation is proper when its polynomial is nonzero | Theorem                      |
| S4 low-order census and cancellation residuals                                                                                             | Computational<br>Certificate |
| S4 numerical Lie-span closure under the declared closure audit                                                                             | Computational<br>Certificate |
| 48 tested direct-channel products are nondegenerate                                                                                        | Computational<br>Observation |
| Matrix-level emergence for general $R_1 = 0$ channels                                                                                      | Research Program             |
| $(R_1, R_2)$ determines full $D$ in represented or dense systems                                                                           | Research Program             |
| Moving accessibility-wall hierarchy                                                                                                        | Research Program             |

The principal failure boundaries are:

Boolean path  $\not\Rightarrow$  nonzero projected product,

nonzero words  $\not\Rightarrow$  nonzero commutator,

and

$$R_1 = R_2 = 0 \not\Rightarrow D = \infty.$$

## 9 Research Boundary

### 9.1 Matrix-level bracket emergence

A matrix-level emergence audit should begin with

$$\mathcal{O}_{R_1} = \{(i, j) : i \neq j, R_1(i, j) = 0\}$$

and separately classify:

1.  $R_2(i, j) = 1$  bracket-emergent channels;
2.  $R_2(i, j) = 0$  with nonzero projected products or words;
3. support paths whose products vanish by image–kernel incidence;
4. pairs with no tested two-step candidate.

Only the first class is a local commutator-layer emergence statement.

## 9.2 Full Lie depth

Recovering exact  $D$  requires either symbolic Lie closure or a certified finite-dimensional closure argument. A numerical certificate should report the filtration ranks, singular values, rank tolerance, and residuals of

$$[X_g, L_a] \in \text{span } \mathcal{L}$$

for every generator and retained Lie-basis element. Cutoff stability alone is not exact closure.

## 9.3 Represented genericity and completion

Whether  $(R_1, R_2)$  determines  $D$  on a checkable represented or dense class is open. Image–kernel incidence geometry, rectangular rank protection, and conditional low-order promotion questions are studied separately [2]. The associated computational atlas and incidence candidates are separate evidence and are not used in the proofs here. In particular, a bridge-level incidence candidate is not automatically a certified accessibility obstruction.

## 9.4 Moving systems

Linearized commutativity and normality constraints, together with pointwise typed registrations on certified commutative-normal samples, are studied in [3]. Moving accessibility fields remain a research problem. Any hierarchy involving  $\Sigma_{R_1}$ ,  $\Sigma_{R_2}$ , or  $\Sigma_D$  belongs to that separate deformation problem. No such hierarchy is proved or assumed here. Cancellation and incidence are static mechanisms, not wall labels.

## 9.5 Higher Hall layers

Jacobi relations and higher Hall monomials can create or remove channels not seen by  $R_2$ . Claims about Jacobi latency, dense-system completion, or a universal finite truncation require independent certificates and are not claimed here.

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# 10 Related Work and Novelty Boundary

Hall bases and free Lie algebras provide the standard organization of commutators and nested commutators [6, 11]. Geometric control theory and structural controllability supply broader Lie-accessibility and graph-generic settings [9, 10]. Zero-pattern and combinatorial matrix theory study which matrix properties are visible from support data [1]. Quivers and path algebras provide a standard language for labelled paths and their algebraic evaluation [12]. Matrix-product survival and rank protection are elementary finite-dimensional linear algebra [8]; incidence language uses standard algebraic geometry [7, 5].

These structural theories motivate the Boolean and path shadows used here, but they do not identify a support path with its value under a concrete matrix or Hall evaluation. The present contribution isolates that evaluation gap through exact signed cancellation, routed/full-word separation, and a local commutator-support boundary.

Support graphs and projected products for represented group elements are separated in [4]. Those products are not identified here with the logarithmic-generator products. The contribution is the exact same-support commutator counterexample, the explicit low-order object separation, and the local bracket-emergence/incidence boundary.

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## 11 Conclusion

Direct support does not determine commutator support. A Boolean two-step path is not a matrix product, a matrix product is not a word sum, a word sum is not a commutator, and low-order absence does not imply infinite-depth inaccessibility.

The exact  $3 \times 3$  example proves that identical generator-indexed support can produce different commutator support and different full Lie depth. The centered scalar proposition identifies one precise local emergence mechanism:

$$R_1(i, j) = 0, \quad R_2(i, j) = 1.$$

For matrix blocks, image–kernel incidence is the missing product datum. The  $S_4$  realization supplies a computational case study of direct, bracket-emergent, cancellation, and cutoff-unreached channels, whereas the 48-product audit is explicitly restricted to already-direct channels.

The resulting scope is deliberately local. General matrix emergence and exact closure in representation-derived systems remain open. Completion from  $(R_1, R_2)$  to  $D$ , and moving accessibility walls, remain research programs.

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## Appendix A Computational Artifacts

The following repository artifacts support the exact counterexample and the computational  $S_4$  case study. The default directory is `experiments/paper5/`; paths are relative to that directory.

| Artifact | Role                                         | Short path                                                         |
|----------|----------------------------------------------|--------------------------------------------------------------------|
| A1       | exact same-support commutator counterexample | <code>validation/exact_support_commutator_counterexample.py</code> |
| A2       | typed $R_1/R_2/C_2^X/W_2^X$ low-order census | <code>validation/low_order_channel_audit.py</code>                 |
| A3       | cutoff Lie-depth and closure audit           | <code>validation/s4_r1_r2_depth.py</code>                          |
| A4       | projected-product cancellation residuals     | <code>validation/path_commutator_cancellation.py</code>            |
| A5       | centered scalar bracket-emergence example    | <code>validation/complement_explosion.py</code>                    |
| A6       | support-set enumeration control              | <code>validation/noncomplement_obstruction_enumeration.py</code>   |
| A7       | direct-channel product nondegeneracy audit   | <code>validation/matrix_nondegeneracy.py</code>                    |

From the repository root, run an artifact as `python experiments/paper5/<short path>`. A1 uses exact integer arithmetic. The  $S_4$  and matrix-logarithm artifacts are numerical and must declare `dtype`, `tolerance`, `generator family`, `logarithm branch`, and `cutoff`. A passing assertion supports only the claim mapped to that artifact.

All listed artifacts are available in the [RIME repository](#).

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